Week 12

Case Study 1

Data for Questions 1 to 5

Clinic Discharge Note Conversion

A regional hospital converts discharge notes written by doctors into a shorter patient-facing language version. The source note may include diagnosis, medicine timing, diet restrictions, warning symptoms, and follow-up dates. The system uses an encoder to read the source note and a decoder to produce the converted note token by token.

In an early prototype, the encoder passes only one final summary vector to the decoder. The converted notes are fluent for short source notes, but long source notes sometimes lose early warnings such as "avoid aspirin" or "return if fever continues." The technical team therefore compares the fixed-summary design with a design where the decoder can look back at multiple encoder states while generating each output token.

Question 1: In the early prototype, what is the main risk of using only one final summary vector?

A) Important source details may be compressed away.

B) The decoder must output every source word unchanged.

C) The encoder is unable to read numerical tokens.

D) The vocabulary becomes empty during generation.

Correct Answer: A

Explanation: A single fixed-length vector can become a bottleneck when the source note is long and contains many details.

Question 2: What does the encoder mainly provide to the decoder in this sequence model?

A) A sorted list of output tokens for the note

B) The final patient label after thresholding

C) Representations derived from the source note

D) A replacement for the training loss function

Correct Answer: C

Explanation: The encoder reads the input sequence and produces representations that summarize or encode its content.

Question 3: Why is the decoder described as producing the converted note token by token?

A) It ignores all tokens generated earlier.

B) Each new token can depend on earlier output tokens.

C) It computes the encoder states after every output only.

D) Every output token must have the same probability.

Correct Answer: B

Explanation: Autoregressive decoding generates one token at a time, usually conditioning on previous generated tokens.

Question 4: Which change most directly helps the decoder recover a warning sentence from early in a long note?

A) Removing all source-token representations

B) Replacing the decoder with a lookup table

C) Using only the last generated token as input

D) Allowing access to relevant encoder states

Correct Answer: D

Explanation: Attention-style access lets the decoder use the encoder state associated with the relevant source part.

Question 5: If a long converted note omits an early allergy warning, what does that suggest about the fixed-summary design?

A) The final summary did not preserve all needed information.

B) The output language has no word order requirement at any step.

C) The source note was processed without any token representations.

D) The decoder cannot produce medical vocabulary for short notes.

Correct Answer: A

Explanation: Missing early content is consistent with information loss caused by compressing the whole input into one vector.

Case Study 2

Data for Questions 6 to 10

Maintenance Ticket Digest

A manufacturing support desk generates a short digest from long maintenance tickets. A ticket may contain the machine ID, fault symptoms, technician notes, replacement parts, and final status. During one decoder step, the model computes alignment scores between the current decoder state and four encoder states representing four ticket sections. Softmax converts the scores into attention weights, and the context vector is computed from the encoder states.

For a particular output token, the attention weights over the four sections are

$$0.10, \quad 0.25, \quad 0.15, \quad 0.50.$$

The four weights correspond to Sections 1, 2, 3, and 4 respectively.

Question 6: What is the role of the attention weights in this decoder step?

A) They replace all encoder states with fixed zero vectors.

B) They set the number of output classes for the digest.

C) They specify how much each section contributes.

D) They sort the ticket sections into alphabetical order.

Correct Answer: C

Explanation: Attention weights determine the relative contribution of each encoder state to the current context.

Question 7: Why is softmax applied to the alignment scores?

A) To convert scores into normalized nonnegative weights.

B) To remove the decoder hidden state from every step.

C) To force the largest raw score to become exactly one.

D) To make all ticket sections contribute with equal weights.

Correct Answer: A

Explanation: Softmax turns raw alignment scores into a distribution of attention weights that sum to one.

Question 8: Which ticket section contributes most to this output token?

A) Section 1

B) Section 2

C) Section 3

D) Section 4

Correct Answer: D

Explanation: Section 4 has the largest attention weight, 0.50.

Question 9: How is the context vector computed after these weights are available?

A) By choosing the first encoder state only

B) By forming a weighted sum of encoder states

C) By averaging decoder outputs from all steps

D) By discarding the highest-weight section

Correct Answer: B

Explanation: The context vector is the weighted combination of encoder states using the attention weights.

Question 10: If the support team expects the replacement-part section to drive this token, what attention pattern would support that interpretation?

A) All four sections receiving negative attention weights

B) The decoder state being removed from the scoring step

C) The replacement-part section receiving the largest weight

D) Every ticket section receiving exactly the same weight

Correct Answer: C

Explanation: A larger attention weight indicates that the corresponding section has stronger influence on the current output token.

Case Study 3

Data for Questions 11 to 15

Procurement Rule Checking

A public procurement office reviews long tender documents. A phrase in one section may refer to eligibility rules, exceptions, or definitions written many pages earlier. The review model uses self-attention so each token representation can compare itself with other token representations in the same document.

For each token, the model forms Query, Key, and Value vectors. Attention scores are computed from query-key dot products, scaled by $\sqrt{d_k}$, passed through softmax, and then used to combine value vectors. Positional encodings are added to token embeddings because the model must also distinguish whether a clause appears before or after another clause.

Question 11: What advantage does self-attention give for checking references across distant clauses?

A) It forces every token to use only its previous neighbor.

B) It removes all long-distance comparisons.

C) It replaces the document with a single class label.

D) It lets tokens compare directly with other tokens.

Correct Answer: D

Explanation: Self-attention allows direct interactions between tokens, including tokens that are far apart in the document.

Question 12: In the query-key comparison, what do the resulting scores represent?

A) The final number of tender documents reviewed

B) Relevance between one token and another token

C) The optimizer's learning-rate value during training

D) The length of the output vocabulary for labels

Correct Answer: B

Explanation: Query-key dot products measure how strongly one token should attend to another.

Question 13: What is the purpose of scaling the dot product by $\sqrt{d_k}$?

A) To keep score magnitudes controlled before softmax

B) To erase the effect of all value vectors in the layer

C) To make positional encodings unnecessary for ordering

D) To force all attention scores to become identical

Correct Answer: A

Explanation: Scaling prevents large dot products from making the softmax distribution too saturated.

Question 14: What role do Value vectors play after attention weights are computed?

A) They choose the batch size for training runs.

B) They remove token-order information from clauses.

C) They provide the content that gets combined.

D) They store only the document identifier value.

Correct Answer: C

Explanation: Attention weights are applied to value vectors to build the output representation.

Question 15: Why are positional encodings needed in this tender-review model?

A) They delete repeated words from the document.

B) They replace softmax in attention scoring.

C) They make every token attend to itself only.

D) They supply information about token order.

Correct Answer: D

Explanation: Self-attention by itself does not encode word order, so positional information must be added.

Case Study 4

Data for Questions 16 to 20

Campus Feedback Review

A university analyzes long feedback reports submitted by students after laboratory courses. Each report contains several paragraphs, each paragraph contains sentences, and each sentence contains words describing equipment, instructions, grading, or safety concerns. The model first builds sentence representations from word representations and then builds a report representation from sentence representations.

The team adds attention at the word level to highlight important phrases inside a sentence and attention at the sentence level to highlight the most informative sentences in the full report. They also test multiple attention heads so different heads can track different evidence types, such as safety terms, assessment complaints, and equipment names.

Question 16: Why is a two-level representation natural for these reports?

A) Reports contain only a single unordered word list.

B) Words form sentences, and sentences form reports.

C) The model must ignore sentence boundaries.

D) Every paragraph has exactly one token.

Correct Answer: B

Explanation: The hierarchy of words within sentences and sentences within reports matches the document structure.

Question 17: What does word-level attention help identify?

A) The number of reports in the dataset split

B) The final optimizer update rule for training

C) The batch size used for report classification

D) Important words or phrases inside a sentence

Correct Answer: D

Explanation: Word-level attention highlights terms inside a sentence that matter for the sentence representation.

Question 18: What does sentence-level attention help identify?

A) Sentences that are most useful for the report decision

B) Characters that should be removed from each word

C) Output tokens generated by a decoder layer

D) The number of attention heads in the layer

Correct Answer: A

Explanation: Sentence-level attention selects the sentences that contribute most to the document-level representation.

Question 19: Why can multiple attention heads be useful in this feedback model?

A) They force all heads to learn identical attention weights.

B) They delete word-level and sentence-level evidence.

C) They allow different evidence patterns to be tracked.

D) They make positional information impossible to use.

Correct Answer: C

Explanation: Different heads can learn to focus on different kinds of relationships or evidence.

Question 20: If sentence attention is almost uniform across a long report, what concern should the team consider?

A) The report contains no sentence representations.

B) The model may not be selecting the most informative sentences.

C) Multi-head attention cannot operate on textual inputs.

D) Word-level attention must always be zero for all words.

Correct Answer: B

Explanation: Nearly uniform sentence attention can indicate weak discrimination among important and routine sentences.